

Milestone 3 Progress Report

Project Name: NeuroFleetX – Urban Power Fleet Management System for Buses

1. Executive Summary

Milestone 3 marks the completion of the core user interface and the end-to-end integration of the fleet management system. Following Milestone 2, the project has transitioned from static designs to a dynamic, data-driven platform, focusing on real-time Passenger and Driver dashboard functionalities.

2. Key Accomplishments

A. Full Passenger Dashboard Development

- **Comprehensive Module Build:** Engineered the entire Passenger Dashboard to provide a seamless user experience.
- **Frontend-Backend Integration:** Successfully linked the dashboard with backend services to enable real-time data fetching for bus schedules and availability.
- **Key Features:**
 - Interactive ticket booking system.
 - Real-time bus tracking and ETA notifications.
 - User account management and trip history.

B. Driver Dashboard & Route Implementation

- **Driver Interface Optimization:** Developed a dedicated workspace for drivers to manage assignments and monitor their shift status.
 - **Live Route Navigation:** Integrated route-tracking features that allow drivers to follow assigned paths while feeding location data back to the system.
 - **Synchronization:** Ensured that the Driver Dashboard communicates directly with the centralized server to update passenger-side maps.
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3. Challenges Overcome

- **Secure Authentication & Access Control:** Encountered and resolved repository access challenges (HTTP 403 Forbidden errors) by implementing **GitHub Personal Access Tokens (PAT)** and **Secure Browser-based Authentication**, ensuring a more secure development environment.

- **Unrelated History Reconciliation:** Faced technical hurdles while merging branches with different commit histories. This was successfully resolved using advanced Git strategies (e.g., --allow-unrelated-histories) to maintain codebase integrity.
 - **Data Flow Synchronization:** Integrating complex React components with the Spring Boot backend required meticulous state management. We successfully implemented a centralized state architecture to ensure data consistency across the Passenger and Driver modules.
 - **Permission & Collaborator Management:** Managed cross-account repository permissions efficiently, enabling a collaborative workflow without disrupting the main master/main branch history.
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4. Technical Stack

- **Frontend:** React.js, Tailwind CSS, State Management (Zustand/Context API).
 - **Backend:** Spring Boot, JPA, RESTful Services.
 - **Version Control:** Git/GitHub with Feature-based Branching.
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5. Conclusion

The project has achieved its primary integration goals for Milestone 3. Both Passenger and Driver modules are now functional and synchronized with the backend. The system is ready for the next phase of performance testing and deployment preparation.